

MANKATO, MN, USA

WMO: 726585

Lat: 44.217N

Lon: 93.917W

Elev: 311

StdP: 97.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14954

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.4	-22.3	-29.6	0.3	-23.5	-27.0	0.3	-21.8	13.8	-10.2	12.5	-11.0	4.0	310	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	32.1	23.2	30.2	22.3	28.6	21.5	25.1	29.7	23.9	28.2	22.8	27.1	6.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.1	28.3	22.4	17.8	27.1	21.3	16.5	25.9	78.2	29.5	72.9	28.3	68.5	27.2	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	10.8	9.3	-26.9	35.0	3.1	1.8	-29.2	36.3	-31.0	37.4	-32.8	38.4	-35.1	39.8		
			-27.4	26.8	2.9	1.3	-29.5	27.7	-31.2	28.4	-32.9	29.1	-35.0	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-9.2	-7.1	0.2	8.2	14.8	20.4	22.4	20.9	17.0	9.5	1.5	-6.1
	DBStd	12.41	7.33	7.09	6.98	5.52	4.92	3.69	3.01	2.95	4.61	5.22	6.21	6.72
	HDD10.0	2328	595	478	312	96	12	0	0	0	4	72	261	499
	HDD18.3	4214	853	711	562	306	133	21	4	9	78	276	504	757
	CDD10.0	1524	0	0	8	42	161	312	383	338	215	58	7	0
	CDD18.3	368	0	0	1	2	22	83	129	88	39	4	0	0
	CDH23.3	2979	0	0	2	32	242	674	1076	592	319	40	1	0
	CDH26.7	859	0	0	0	8	83	216	328	134	81	10	0	0
Wind	WSAvg	4.6	4.9	5.0	5.0	5.4	5.0	4.3	3.6	3.4	4.1	4.8	5.0	4.7
Precipitation	PrecAvg	860	26	22	51	76	102	130	110	119	83	65	45	30
	PrecMax	1114	71	54	100	147	186	214	188	248	253	156	117	96
	PrecMin	583	3	2	7	17	37	76	43	40	19	16	5	6
	PrecStd	124	16	13	23	35	43	44	46	46	53	40	30	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	11.7	21.2	27.4	32.7	33.8	33.7	32.2	32.1	27.8	20.3	11.8
		MCWB	3.6	8.4	14.3	16.5	20.1	23.0	24.3	25.1	21.4	18.5	13.2	9.0
	2%	DB	3.8	7.3	17.0	22.6	28.8	31.3	32.1	29.3	28.9	23.1	16.3	7.2
		MCWB	1.7	4.7	12.2	13.5	17.8	21.9	24.7	23.5	21.1	16.0	11.1	4.6
	5%	DB	2.3	4.1	12.9	19.7	26.1	28.9	30.0	27.8	27.2	20.9	12.8	3.8
		MCWB	0.9	2.2	9.2	12.1	16.7	20.5	23.4	22.1	19.4	14.6	8.7	1.9
	10%	DB	0.9	2.4	9.3	17.2	22.8	27.3	28.0	27.0	25.0	17.6	10.1	2.4
		MCWB	-0.3	0.8	6.4	10.4	14.8	19.4	21.9	21.1	18.8	12.2	6.7	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	8.5	16.1	17.4	22.1	24.9	27.2	26.4	24.1	20.2	14.6	9.6
		MCDB	6.2	11.3	19.1	25.0	28.5	30.9	31.6	30.0	28.7	24.7	18.9	11.7
	2%	WB	2.0	4.9	12.9	15.0	19.7	23.3	25.5	24.5	22.5	17.1	11.7	4.5
		MCDB	3.5	7.2	16.2	20.8	26.1	28.8	30.4	28.3	26.9	21.7	15.5	6.7
	5%	WB	0.8	2.6	9.9	13.0	17.9	21.9	24.2	23.3	21.1	15.0	9.1	2.3
		MCDB	2.1	4.4	13.1	18.4	23.4	27.6	28.7	27.0	25.3	19.5	12.6	3.9
	10%	WB	-0.5	0.8	6.7	11.0	16.4	20.5	23.1	22.1	19.7	12.9	6.9	0.7
		MCDB	0.5	2.4	9.8	16.2	22.0	25.7	27.3	25.7	23.6	17.2	9.9	2.1
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	8.6	10.2	10.3	9.7	9.6	9.5	10.8	9.9	8.4	7.5
		MCDBR	8.7	9.4	13.0	15.2	14.6	12.2	10.8	10.4	13.0	14.3	12.7	9.9
		MCWBR	7.3	7.7	8.9	8.7	7.0	5.9	6.2	6.1	6.7	8.1	8.4	8.1
	5% WB	MCDBR	8.5	8.9	12.4	13.0	11.9	10.7	9.9	9.5	11.0	12.1	11.9	9.3
		MCWBR	7.3	7.3	8.9	8.4	6.5	6.2	6.3	6.3	6.7	7.9	8.4	7.7
Clear-Sky Solar Irradiance	taub	0.265	0.286	0.311	0.368	0.390	0.406	0.413	0.407	0.374	0.323	0.288	0.266	
	taud	2.449	2.382	2.434	2.320	2.319	2.319	2.298	2.299	2.361	2.506	2.520	2.460	
	Ebn at Noon	876	910	930	895	881	865	854	845	847	858	841	837	
	Edh at Noon	70	90	99	121	125	126	127	122	106	80	66	64	
All-Sky Solar Radiation	RadAvg	1.78	2.85	3.72	4.60	5.38	6.01	6.41	5.34	4.23	2.75	1.78	1.31	
	RadStd	0.21	0.31	0.37	0.48	0.42	0.49	0.26	0.39	0.30	0.36	0.17	0.12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page